

Computer Science & Engineering



LABORATORY NOTE BOOK

MAKAUT EVEN SEMESTER 2025 - 26

PAPER NAME: DATABASE MANAGEMENT SYSTEMS LAB

PAPER CODE: PCC – CS691

STUDENT NAME: _____

ROLL NO.: _____ . REGN. NO.: _____ .

YEAR.: _____ . SEMESTER.: _____ .

DEPARTMENT.: _____ . SESSION.: _____ .



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“LIST OF ASSIGNMENT/EXPERIMENT SUBMISSION DETAILS”

SL. NO.	ASSIGNMENT / EXPERIMENT NAME	DATE OF ASSIGNMENT / EXPERIMENT DONE	DATE OF SUBMISSION	CHECKED BY	REMARKS (ANY DEVIATION REGARDING SUBMISSION DATES, CONTENT, FORMAT, ETC)

OBSERVATIONS / COMMENTS ON THE OVERALL PERFORMANCE:

Signature in full with date
Faculty / Technical Assistant

Assignment-1

Create these two tables, EMP and DEPT in your own LIVE SQL account.

EMP

EMPNO	EMPNAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
7369	Smith	Clerk	7902	17/12/2015	50000	1000	20
7499	Allen	salesman	7698	20/02/2016	42000	1000	30
7521	Ward	salesman	7698	22/02/2016	43000	1000	30
7566	Jones	Maneger	7839	02/04/2018	75000	0	20
7654	Martin	salesman	7696	28/09/2017	45000	500	30
7698	Blake	Maneger	7839	01/05/2017	80000	0	30
7782	Clark	Maneger	7839	09/06/2018	78000	0	10
7788	Scott	Analyst	7566	19/04/2023	125000	0	20
7839	King	President		17/11/2017	235000	0	40
7844	Turner	salesman	7698	08/09/2015	46000	0	30

DEPT

DEPTNO	DEPTNAME	LOC
10	ACCOUNTING	NEW YORK
20	RESEARCH	DALLAS
30	SALES	CHICAGO
40	OPERATIONS	BOSTON

Statement 1

```
Create Table DEPT (
DEPTNO Number(5) Primary Key,
DEPTNAME Varchar2(15) Not Null,
LOC Varchar2(20))
```

Table created.

Statement 2

```
DESC DEPT
TABLE DEPT
```

Column	Null?	Type
DEPTNO	NOT NULL	NUMBER(5,0)
DEPTNAME	NOT NULL	VARCHAR2(15)
LOC	-	VARCHAR2(20)

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Statement 3

```
Create Table EMP
EMPNO Number(5) Primary Key,
EMPNAME Varchar2(15) Not Null,
JOB Varchar2(15),
MGR Number(5),
HIREDATE Date,
SAL Number(10,2),
COMM Number(10,2),
DEPTNO Number(5) References DEPT(DEPTNO))
```

Table created.

Statement 4

```
DESC EMP
TABLE EMP
```

Column	Null?	Type
EMPNO	NOT NULL	NUMBER(5,0)
EMPNAME	NOT NULL	VARCHAR2(15)
JOB	-	VARCHAR2(15)
MGR	-	NUMBER(5,0)
HIREDATE	-	DATE
SAL	-	NUMBER(10,2)
COMM	-	NUMBER(10,2)
DEPTNO	-	NUMBER(5,0)

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Statement 5

```
INSERT INTO DEPT VALUES (10, 'ACCOUNTING', 'NEW YORK')
```

1 row(s) inserted.

Statement 6

```
INSERT INTO DEPT VALUES (30, 'SALES', 'CHICAGO')
```

1 row(s) inserted.

Statement 7

```
SELECT * FROM DEPT
```

DEPTNO	DEPTNAME	LOC
10	ACCOUNTING	NEW YORK
30	SALES	CHICAGO

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Statement 8

```
Insert into EMP Values(7369,'Smith','Clerk',7902,'17-DEC-2015',50000,1000,20)
```

Statement 9

```
Insert into EMP Values(7698,'Blake','Maneger',7839,'01-MAY-2017',80000,0,30)
```

1 row(s) inserted.

Statement 10

```
SELECT * FROM EMP
```

EMPNO	EMPNAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
7369	Smith	Clerk	7902	17-DEC-15	50000	1000	20
7698	Blake	Manager	7839	01-MAY-17	80000	0	30

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2 rows selected.

Assignment-2

Using EMP and DEPT databases solve some queries:

1. List the employee belonging to the department 20.

```
Select EMPNAME from EMP where DEPTNO=20
```

EMPNAME
Smith
Jones
Scott

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3 rows selected.

2. List the name and salary of the employee whose salary is more than 50000.

```
Select EMPNAME, SAL from EMP where SAL>50000
```

EMPNAME	SAL
Jones	75000
Blake	80000
Clark	78000
Scott	125000
King	235000

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5 rows selected.

3. List the employee number and name of managers.

Select EMPNO, EMPNAME from EMP where JOB='Manager'

EMPNO	EMPNAME
7566	Jones
7698	Blake
7782	Clark

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3 rows selected.

4. List the names of the clerk working in the department no 20.

Select EMPNAME from EMP where JOB='Clerk' and DEPTNO=20

EMPNAME
Smith

Download CSV

5. List the name of analyst and salesman.

Select EMPNAME from EMP where JOB='Analyst' or
JOB='salesman'

EMPNAME
Allen
Ward
Martin
Scott
Turner

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6. List the details of the employee who have joined before the end of the September'2020.

```
Select * from EMP where  
HIREDATE<to_date('30/09/2020','DD/MM/YYYY')
```

EMPNO	EMPNAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
7369	Smith	Clerk	7902	17-DEC-15	50000	1000	20
7499	Allen	salesman	7698	20-FEB-16	42000	1000	30
7521	Ward	salesman	7698	22-FEB-16	43000	1000	30
7566	Jones	Manager	7839	02-APR-18	75000	0	20
7654	Martin	salesman	7696	28-SEP-17	45000	500	30
7698	Blake	Manager	7839	01-MAY-17	80000	0	30
7782	Clark	Manager	7839	09-JUN-18	78000	0	10
7839	King	President	-	17-NOV-17	235000	0	40
7844	Turner	salesman	7698	08-SEP-15	46000	0	30

Download CSV

9 rows selected.

7. List the names of employee who are not managers.

```
Select EMPNAME from EMP where JOB!='Manager'
```

EMPNAME
Smith
Allen
Ward
Jones
Martin
Blake
Clark
Scott
Turner
King

Download CSV

Assignment-3

1. List the name of the employee whose employee number are 7369, 7839, 7934, 7788.

Select EMPNAME from EMP where EMPNO in (7369, 7839, 7934, 7788)

EMPNAME
Smith
Scott
King

Download CSV

3 rows selected

2. List the employee details not belonging to the department 10, 30, 40.

Select * from EMP where DEPTNO not in (10, 30, 40)

EMPNO	EMPNAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
7369	Smith	Clerk	7902	17-DEC-15	50000	1000	20
7566	Jones	Maneger	7839	02-APR-18	75000	0	20
7788	Scott	Analyst	7566	19-APR-23	125000	0	20

Download CSV

3 rows selected.

3. List employee names who have joined before 30th June'2018 and after December'2018.

```
Select EMPNAME from EMP where  
HIREDATE<to_date('30/06/2018','DD/MM/YYYY') or  
HIREDATE>to_date('31/12/2018','DD/MM/YYYY')
```

EMPNAME
Smith
Allen
Ward
Jones
Martin
Blake
Clark
Scott
King
Turner

Download CSV

4. List the different jobs available in the emp table.

```
Select distinct JOB from EMP
```

JOB
Maneger
President
salesman
Clerk
Analyst

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5 rows selected.

5. List the employee names ending with an 'S'.

```
Select EMPNAME from EMP where EMPNAME like '%s'
```

EMPNAME
Jones

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6. List the employee names having 'i' as the second character.

```
Select EMPNAME from EMP where EMPNAME like '_i%'
```

EMPNAME
King

Download CSV

7. List the names of employee, who are more than 5 years old in the organization.

```
Select EMPNAME from EMP where  
months_between(SYSDATE, HIREDATE) > 60
```

EMPNAME
Smith
Allen
Ward
Jones
Martin
Blake
Clark
King
Turner

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